

Model Assumptions and Limitations

(Documented directly in Notebook Commenting for source code file Machine Learning Model.ipynb)

Modelling Grain and Target

The authoritative modelling grain is **one row per student_deidentified_hash**.

The executed dataset contains:

- 973,770 fact rows;
- 973,770 distinct students;
- 973,770 distinct enrolments;
- zero NULL student hashes;
- zero NULL enrolment hashes;
- zero NULL target values; and
- zero students with more than one fact row.

One eligible fact row per student is therefore confirmed, and no eligibility rule or deduplication is required.

The supervised target is:

is_twelve_month_student_attrition

and is treated as the sole supervised label.

Feature-Set Assumption

Phase 2 consumes the **21 approved Phase 1 features unchanged**.

No feature is rediscovered or added during model training. The numeric and categorical feature lists must reconstruct the approved feature list exactly, otherwise execution stops.

Train, Validation and Test Data

The approved split proportions are:

- 70% training;
- 15% validation;
- 15% testing.

The random seed is 42.

The implementation assigns students deterministically into 100 hash buckets:

- 0–69 — training;

- 70–84 — validation;
- 85–99 — testing.

Every split must be non-empty, contain both target classes and contain one row per student. No student may occur in more than one split.

Preprocessing Assumptions

Preprocessing stages are fitted on the **training data only**. Nothing is fitted on validation or test data.

Numeric features are:

- cast to double; and
- imputed using the median.

The median values are learned from the training split only.

Categorical features are:

- cast to string;
- indexed using StringIndexer;
- configured with handleInvalid="keep";
- one-hot encoded with handleInvalid="keep".

Unseen and invalid categories are therefore placed into a dedicated bucket rather than dropped.

No scaling is applied because the approved Random Forest does not require scaling.

Class Imbalance

The executed target distribution contains:

- 53,694 positive attrition cases — 5.51%;
- 920,076 negative cases — 94.49%;
- approximately 17:1 negative-to-positive imbalance.

Class weighting is applied when the positive class represents less than 40% of the training data.

The weights are calculated from the **training split only**, so validation and test information does not influence them.

The documented weighting formula is:

$\text{weight}(\text{class } c) = \text{training_rows} / (2 \times \text{training_rows_in_class_c})$

Each class therefore contributes the same total weight.

Class weighting recentres the predicted probabilities around 0.50.

Random Forest Configuration

The Random Forest uses:

- numTrees = 100
- maxDepth = 7
- minInstancesPerNode = 20
- featureSubsetStrategy = "sqrt"
- seed = 42
- the class-weight column when class weighting is required.

The notebook states that this configuration is used without model comparison, competing feature subsets, alternative parameter configurations or hyperparameter search.

Prediction Threshold

The approved risk threshold is **0.50**.

The positive-class probability is used to derive the predicted class and Phase 3 risk flag:

- probability $\geq 0.50 \rightarrow$ true;
- probability $< 0.50 \rightarrow$ false.

The threshold is applied to the **unrounded probability**. The risk flag is not derived from the rounded percentage.

Evaluation Assumptions

Recall, precision and F1 are reported for the **positive attrition class** rather than as weighted averages.

The notebook states that a weighted average over the approximately 94.5% negative class would be dominated by the majority class and would misrepresent the primary metric.

Recall is treated as the primary metric.

ROC-AUC is calculated from the continuous positive-class probability and is therefore threshold-independent.

The model is also required to produce non-trivial predictions:

- probabilities cannot all be constant;
- probability standard deviation must be greater than zero; and
- both positive and negative predictions must be produced.

Known Limitation L-1 — Field of Education Is Decoupled from the Target

The attrition Beta parameters are selected using each fact row's own broad_field_of_education.

That field is never written to the fact table.

The course dimension's field of education is seeded from a different fact row and coincides with the true value on only approximately **0.16% of rows**.

Known Limitation L-2 — The 21 Features Are Close to Flat Against the Target

The student-attrition base rate is approximately **5.51%**.

The widest spread against the target for any feature is documented as **1.16 percentage points**, with most of that spread attributed to sampling noise.

The notebook therefore states that:

- ROC-AUC near 0.5 is expected; and
- precision near the base rate is expected.

This is documented as the correct outcome for the synthetic data rather than a defect in the workflow and is not considered a reason to alter the approved features, model parameters or threshold.

Known Limitation L-3 — Weighted Probabilities Are Not Calibrated

Class weighting recentres probabilities around 0.50 so that the approved threshold sits at the decision boundary.

As a result, the model output is a **relative ranking score rather than a likelihood of attrition**.

The notebook later states this limitation more explicitly: a stored attrition_risk_percentage of 52 does **not** mean that the student has a 52% chance of attrition when the true base rate is 5.51%.

Any application surface presenting the value to staff should describe it as a **relative risk score**.

Inference Population Limitation

The inference population is **every valid student row in the Phase 1 canonical modelling projection**.

This includes training, validation and test rows because the application requires a current risk result for every valid student.

The notebook explicitly states that rows used during training are therefore scored optimistically. This affects the stored percentages, but not the Phase 2 validation and test metrics, which are measured on untouched data.

Rows with an unknown outcome are also scored even though they cannot be used for training.

Preprocessing statistics are not refitted during inference. Missing numeric values use the training median and unseen categories use the reserved category bucket.

Interpretation of Stored Risk Output

attrition_risk_probability represents the model's positive-class probability before publication.

attrition_risk_percentage is this value multiplied by 100.

attrition_risk_flag applies the 0.50 threshold to the unrounded probability.

Because the Random Forest is class weighted, the resulting risk percentage must be interpreted as a **relative risk/ranking score, not a calibrated probability of student attrition**.

Generated using OpenAI with prompt:

Hello, I have attached the source code file for the Machine Learning Model.ipynb. I need you to read through this file and extract the Model Assumptions and Limitations that are already documented in the source code.

Investigate the notebook comments, markdown sections, validation checks and documented limitations that relate to how the model works and how its results should be interpreted.

- Include the main assumptions made when preparing, training and evaluating the model.
- Include any limitations of the model or dataset that are directly documented in the notebook.
- Include any important evidence or results in the notebook that support these assumptions and limitations.
- Only use information and reasoning that is directly documented in the attached notebook file. Do not add external machine learning reasoning or create new assumptions or limitations that are not present in the source code.
- Summarise and organise the extracted information so it can be copied into development documentation under the heading "Model Assumptions and Limitations".

If an assumption or limitation does not have a documented explanation in the notebook, state this rather than creating one.